

Conservative policy Iteration

In the infinite-horizon discounted setting with unknown model, conservative policy iteration aims to

- iteratively evaluate the policy, using supervised learning, and improve the policy, and
- ensure monotonic improvement by keeping the policy updates incremental.

Idea: Applying performance difference lemma on π_{t+1} and π_t , we obtain

$$V^{\pi_{t+1}}(s_0) - V^{\pi_t}(s_0) = \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_{t+1}}} [A^{\pi_t}(s, \pi_{t+1}(s))] \cdot \begin{matrix} \nearrow \text{if } \geq 0 \\ \downarrow \text{improvement} \end{matrix}$$

incremental policy update

$$\text{If } \pi_t \approx \pi_{t+1} \Rightarrow \bar{\rho}_{s_0}^{\pi_t} \approx \bar{\rho}_{s_0}^{\pi_{t+1}}$$

$$\Rightarrow \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_t}} [A^{\pi_t}(s, \pi_{t+1}(s))] \approx \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_{t+1}}} [A^{\pi_t}(s, \pi_{t+1}(s))]$$

$$\Rightarrow \max_{\pi_{t+1}} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_t}} [A^{\pi_t}(s, \pi_{t+1}(s))] \approx \max_{\pi_{t+1}} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_{t+1}}} [A^{\pi_t}(s, \pi_{t+1}(s))]$$

instead of $\bar{\rho}_{s_0}^{\pi_{t+1}}$

Therefore, by maximizing $\mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_t}} [A^{\pi_t}(s, \pi_{t+1}(s))]$, we can hope to maximize $V^{\pi_{t+1}}(s_0) - V^{\pi_t}(s_0)$ and improve the policy.

Note: With function approximation for Q or A and under certain conditions for the regression problem, we have the following:

$$\mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_t}} [A^{\pi_t}(s, \pi_{t+1}(s))] \geq \max_a \hat{A}^{\pi_t}(s, a) \cdot \max_{\pi} \mathbb{E}_{s \sim \bar{\rho}_{s_0}^{\pi_t}} [A^{\pi_t}(s, \pi(s))] - O(\sqrt{\delta})$$

bound on generalization error, e.g., $\delta = \frac{C}{\sqrt{N}}$

Iterative method:

- Initialize $\pi_0 : \mathcal{S} \rightarrow \Delta(\mathcal{A})$

- For $t = 0, 1, 2, \dots, T-1$:

supervised learning for policy evaluation

- Evaluate π_t by collecting a data set $\{(\underbrace{x^i = (s^i, a^i)}_{\text{roll in}}, \underbrace{y^i}_{\text{roll out}})\}_{i=1}^N$ and estimating $\hat{Q}^\pi$ using
$$\hat{Q}^{\pi_t} = \arg \min_{Q \in \mathcal{Q}} \sum_{i=1}^N (Q(s^i, a^i) - y^i)^2$$

policy improvement

- Improve policy by
$$\bar{\pi}(s) = \arg \max_{a \in \mathcal{A}} \hat{Q}^{\pi_t}(s, a) \quad \forall s \in \mathcal{S}$$
- Apply incremental policy update
$$\pi_{t+1}(a|s) = \underbrace{(1-\alpha)\pi_t(a|s) + \alpha\bar{\pi}(a|s)}_{\text{convex combination of two policies}}$$
$$\downarrow \alpha \in [0, 1] \quad \forall s \in \mathcal{S}, \forall a \in \mathcal{A}$$

- Return π_T

Note: In the policy update step, the new policy π_{t+1} acts as

$$\begin{cases} \pi_t & \text{w.p. } 1-\alpha \\ \bar{\pi} & \text{w.p. } \alpha \end{cases}, \text{ where } \alpha \in [0, 1].$$

Monotonic Improvement in Conservative Policy Iteration

We start by bounding the policy update, i.e., the difference between the policies from two consecutive iterations.

$$\begin{aligned} \|\pi_{t+1}(\cdot|s) - \pi_t(\cdot|s)\|_1 &\stackrel{\text{definition of } \pi_{t+1}}{=} \|(1-\alpha)\pi_t(\cdot|s) + \alpha\bar{\pi}(\cdot|s) - \pi_t(\cdot|s)\|_1 \\ &= \|- \alpha\pi_t(\cdot|s) + \alpha\bar{\pi}(\cdot|s)\|_1 \end{aligned}$$

$$\stackrel{\text{triangle inequality}}{\leq} \alpha \|\pi_t(\cdot|s)\|_1 + \alpha \|\bar{\pi}(\cdot|s)\|_1$$

$$\stackrel{\text{policy being a probability distribution}}{\leq} 2\alpha$$

incremental policy update

Lemma: Consider two policies $\pi: S \rightarrow \Delta(A)$ and $\pi': S \rightarrow \Delta(A)$

such that $\|\pi(\cdot|s) - \pi'(\cdot|s)\|_1 \leq \delta$. It holds that

$$\|\bar{P}_{\mu_0}^{\pi} - \bar{P}_{\mu_0}^{\pi'}\|_1 \leq \frac{\gamma\delta}{1-\gamma} \quad \delta = 2\alpha \text{ in case of conservative policy iteration}$$

Proof: Recall the definition of normalized state occupancy measure:

$$\bar{P}_{\mu_0}^{\pi}(s) = (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \underbrace{P_t^{\pi}(s)}_{P_t^{\pi}(s)} = P(s_t = s | \mu_0, P, \pi).$$

$$\text{Let } P_t^{\pi}(s) = P(s_t = s | \mu_0, P, \pi).$$

$$P_t^\pi(s') - P_t^{\pi'}(s') \stackrel{\text{Markov property}}{=} \sum_{s \in \mathcal{S}} P_{t-1}^\pi(s) \sum_{a \in \mathcal{A}} \pi(a|s) P(s'|s, a) - \sum_{s \in \mathcal{S}} P_{t-1}^{\pi'}(s) \sum_{a \in \mathcal{A}} \pi'(a|s) P(s'|s, a)$$

$$\stackrel{\text{rearranging terms}}{=} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} (P_{t-1}^\pi(s) \pi(a|s) - P_{t-1}^{\pi'}(s) \pi'(a|s)) P(s'|s, a)$$

$$\stackrel{\text{adding and subtracting}}{=} \sum_{s \in \mathcal{S}} \sum_{a \in \mathcal{A}} \left(P_{t-1}^\pi(s) \pi(a|s) - \underbrace{P_{t-1}^\pi(s) \pi'(a|s)} + \underbrace{P_{t-1}^\pi(s) \pi'(a|s) - P_{t-1}^{\pi'}(s) \pi'(a|s)} \right) P(s'|s, a)$$

$$\stackrel{\text{rearranging terms}}{=} \sum_{s \in \mathcal{S}} P_{t-1}^\pi(s) \sum_{a \in \mathcal{A}} (\pi(a|s) - \pi'(a|s)) P(s'|s, a) + \sum_{s \in \mathcal{S}} (P_{t-1}^\pi(s) - P_{t-1}^{\pi'}(s)) \sum_{a \in \mathcal{A}} \pi'(a|s) P(s'|s, a)$$

$$\xrightarrow{\quad} |P_t^\pi(s') - P_t^{\pi'}(s')| \stackrel{\text{triangle inequality}}{\leq} \sum_{s \in \mathcal{S}} P_{t-1}^\pi(s) \sum_{a \in \mathcal{A}} |\pi(a|s) - \pi'(a|s)| P(s'|s, a) + \sum_{s \in \mathcal{S}} |P_{t-1}^\pi(s) - P_{t-1}^{\pi'}(s)| \sum_{a \in \mathcal{A}} \pi'(a|s) P(s'|s, a)$$

$$\xrightarrow{\quad} \sum_{s' \in \mathcal{S}} |P_t^\pi(s') - P_t^{\pi'}(s')| \leq \sum_{s \in \mathcal{S}} P_{t-1}^\pi(s) \sum_{a \in \mathcal{A}} |\pi(a|s) - \pi'(a|s)| \underbrace{\sum_{s' \in \mathcal{S}} P(s'|s, a)}_1 + \sum_{s \in \mathcal{S}} |P_{t-1}^\pi(s) - P_{t-1}^{\pi'}(s)| \underbrace{\sum_{s' \in \mathcal{S}} \sum_{a \in \mathcal{A}} \pi'(a|s) P(s'|s, a)}_1$$

definition of L1 norm $\leftarrow$
$$= \sum_{s \in \mathcal{S}} \overbrace{p_{t-1}^{\pi}(s)}^1 \underbrace{\| \pi(\cdot|s) - \pi'(\cdot|s) \|_1}_{\leq \delta}$$

$$+ \sum_{s \in \mathcal{S}} |p_{t-1}^{\pi}(s) - p_{t-1}^{\pi'}(s)|$$

assumption $\leftarrow$
$$\leq \delta + \sum_{s \in \mathcal{S}} |p_{t-1}^{\pi}(s) - p_{t-1}^{\pi'}(s)|$$

repeat the argument $\leftarrow$
$$\leq \delta + \delta + \sum_{s \in \mathcal{S}} |p_{t-2}^{\pi}(s) - p_{t-2}^{\pi'}(s)|$$

after total of t iterations $\leftarrow$
$$\vdots$$

$$\leq \delta t$$

Now, we can bound the difference between the occupancy measures.

definition of $\bar{\rho}$ $\leftarrow$
$$\| \bar{\rho}_{\mu_0}^{\pi} - \bar{\rho}_{\mu_0}^{\pi'} \|_1 = \sum_{s' \in \mathcal{S}} \left| (1-\gamma) \sum_{t=0}^{\infty} \gamma^t p_t^{\pi}(s') \right.$$

$$\left. - (1-\gamma) \sum_{t=0}^{\infty} \gamma^t p_t^{\pi'}(s') \right|$$

triangle inequality $\leftarrow$
$$\leq (1-\gamma) \sum_{t=0}^{\infty} \gamma^t \underbrace{\sum_{s' \in \mathcal{S}} |p_t^{\pi}(s') - p_t^{\pi'}(s')|}_{\leq \delta t}$$

arithmetic-geometric series $\leftarrow$
$$\leq (1-\gamma) \delta \sum_{t=0}^{\infty} \gamma^t t$$

$$= (1-\gamma) \delta \frac{\gamma}{(1-\gamma)^2}$$

$$= \frac{\gamma \delta}{1-\gamma}$$



Theorem [Monotonic Improvement of CPI]: If the regression problem works perfectly,
the conservative policy iteration algorithm guarantees monotonic improvement,

$$V^{\pi_{t+1}}(\mu_0) - V^{\pi_t}(\mu_0) = \mathbb{E}_{s \sim \mu_0} [V^{\pi_{t+1}}(s) - V^{\pi_t}(s)] \geq \frac{1-\gamma}{8\gamma R_{\max}} \Delta_t^2 \geq 0$$

with step size $\alpha = \frac{(1-\gamma)^2 \Delta_t}{4\gamma R_{\max}}$, where $\Delta_t = \mathbb{E}_{s \sim \bar{\rho}^{\pi_t}} [A^{\pi_t}(s, \bar{\pi}(s))]$.

$\rightarrow 0 \leq R(s, a) \leq R_{\max}$

Proof:

Performance difference lemma

$$\mathbb{E}_{s \sim \mu_0} [V^{\pi_{t+1}}(s) - V^{\pi_t}(s)] = \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^{\pi_{t+1}}} [\mathbb{E}_{a \sim \pi_{t+1}(s)} [A^{\pi_t}(s, a)]]$$

definition of $\pi_{t+1} \leftarrow$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^{\pi_{t+1}}} \left[(1-\alpha) \mathbb{E}_{a \sim \pi_t(s)} [A^{\pi_t}(s, a)] + \alpha \mathbb{E}_{a \sim \bar{\pi}(s)} [A^{\pi_t}(s, a)] \right]$$

definition of A^{π_t} + determinism of $\bar{\pi}$ $\leftarrow$

$$= \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}^{\pi_{t+1}}} [0 + \alpha A^{\pi_t}(s, \bar{\pi}(s))]$$

adding and subtracting $\leftarrow$

$$= \frac{\alpha}{1-\gamma} \left(\mathbb{E}_{s \sim \bar{\rho}^{\pi_{t+1}}} [A^{\pi_t}(s, \bar{\pi}(s))] - \mathbb{E}_{s \sim \bar{\rho}^{\pi_t}} [A^{\pi_t}(s, \bar{\pi}(s))] + \mathbb{E}_{s \sim \bar{\rho}^{\pi_t}} [A^{\pi_t}(s, \bar{\pi}(s))] \right)$$

definition of expectation $\leftarrow$

$$= \frac{\alpha}{1-\gamma} \left(\sum_{s \in \mathcal{S}} (\bar{\rho}^{\pi_{t+1}}(s) - \bar{\rho}^{\pi_t}(s)) A^{\pi_t}(s, \bar{\pi}(s)) + \mathbb{E}_{s \sim \bar{\rho}^{\pi_t}} [A^{\pi_t}(s, \bar{\pi}(s))] \right)$$

$$\overset{\substack{\alpha \geq -|x| \\ \text{definition of } \Delta_t}}{\geq} \frac{\alpha}{1-\gamma} \left(- \left| \sum_{s \in \mathcal{S}} (\bar{\rho}_{\mu_0}^{\pi_{t+1}}(s) - \bar{\rho}_{\mu_0}^{\pi_t}(s)) A^{\pi_t}(s, \bar{\pi}(s)) \right| + \Delta_t \right)$$

$$\overset{\text{triangle inequality}}{\geq} \frac{\alpha}{1-\gamma} \left(- \sum_{s \in \mathcal{S}} |(\bar{\rho}_{\mu_0}^{\pi_{t+1}}(s) - \bar{\rho}_{\mu_0}^{\pi_t}(s))| \underbrace{|A^{\pi_t}(s, \bar{\pi}(s))|}_{\leq \frac{R_{\max}}{1-\gamma}} + \Delta_t \right)$$

$$\left[A^{\pi}(s, a) = Q^{\pi}(s, a) - V^{\pi}(s) = R(s, a) - R(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim p(\cdot | s, a)} [V^{\pi}(s')] - \gamma \mathbb{E}_{s' \sim p(\cdot | s, \pi(s))} [V^{\pi}(s')] \leq \sum_{t=0}^{\infty} \gamma^t R_{\max} \right]$$

$$\geq \frac{\alpha}{1-\gamma} \left(- \frac{R_{\max}}{1-\gamma} \sum_{s \in \mathcal{S}} |(\bar{\rho}_{\mu_0}^{\pi_{t+1}}(s) - \bar{\rho}_{\mu_0}^{\pi_t}(s))| + \Delta_t \right)$$

$$\overset{\text{definition of L1 norm}}{=} \frac{\alpha}{1-\gamma} \left(- \frac{R_{\max}}{1-\gamma} \|\bar{\rho}_{\mu_0}^{\pi_{t+1}} - \bar{\rho}_{\mu_0}^{\pi_t}\|_1 + \Delta_t \right)$$

$$\overset{\text{from the previous lemma}}{\geq} \frac{\alpha}{1-\gamma} \left(- \frac{R_{\max}}{1-\gamma} \frac{\gamma(2\alpha)}{1-\gamma} + \Delta_t \right)$$

$$= \frac{1}{1-\gamma} \left(- \underbrace{\frac{2\gamma R_{\max}}{(1-\gamma)^2} \alpha^2 + \Delta_t \alpha}_{B(\alpha)} \right)$$

$B(\alpha) \rightarrow$ concave function in α



To maximize the amount of policy improvement, we pick the step size such that the lower bound above is maximized.

$$\Delta_t \geq 0 \Rightarrow \alpha \geq 0$$

$$\text{if } \gamma \geq \frac{1}{5} \Rightarrow \alpha \leq 1$$

$$\longrightarrow \nabla_{\alpha} B(\alpha) = - \frac{4\gamma R_{\max}}{(1-\gamma)^2} \alpha + \Delta_t = 0 \Rightarrow \alpha = \frac{(1-\gamma)^2 \Delta_t}{4\gamma R_{\max}}$$

$$\longrightarrow \mathbb{E}_{s \sim \mu_0} [V^{\pi_{t+1}}(s) - V^{\pi_t}(s)] \geq \frac{1-\gamma}{8\gamma R_{\max}} \Delta_t^2 \geq 0$$

monotonic improvement

Near-optimality of Conservative Policy Iteration

Suppose we run the CPI algorithm until the following termination criterion holds:

$$\Delta_t = \mathbb{E}_{s \sim \bar{p}_{\mu_0}^{\pi_t}} [A^{\pi_t}(s, \bar{\pi}(s))] \leq \varepsilon.$$

Theorem [Near-optimality of CPI]: The CPI algorithm with the above termination criterion terminates in at most $\left\lceil \frac{8\gamma R_{\max}^2}{(1-\gamma)^2 \varepsilon^2} \right\rceil$ iterations and outputs policy π_T satisfying the near-optimality guarantee

$$V(\mu_0)^* - V(\mu_0)^{\pi_T} \leq \frac{\varepsilon}{(1-\gamma)^2} \left(\max_{s \in \mathcal{S}} \frac{\bar{p}_{\mu_0}^{\pi^*}(s)}{\mu_0(s)} \right)$$

if $\max_{s \in \mathcal{S}} \frac{\bar{p}_{\mu_0}^{\pi^*}(s)}{\mu_0(s)} < \infty$.

Proof: With step size selected by $\alpha = \frac{(1-\gamma)^2 \Delta_t}{4\gamma R_{\max}}$, we saw that monotonic improvement is guaranteed:

$$\mathbb{E}_{s \sim \mu_0} [V^{\pi_{t+1}}(s) - V^{\pi_t}(s)] = V^{\pi_{t+1}}(\mu_0) - V^{\pi_t}(\mu_0) \geq \frac{1-\gamma}{8\gamma R_{\max}} \Delta_t^2 \geq 0.$$

If the algorithm does not terminate at time $t-1$, it means that

$\Delta_i > \varepsilon$ in all iterations until and including $t-1$.

—————→

$$V^{\pi_t}(\mathcal{M}_0) \geq V^{\pi_{t-1}}(\mathcal{M}_0) + \frac{1-\gamma}{8\gamma R_{\max}} \epsilon^2$$

repeat the argument $\leftarrow$

$$\geq V^{\pi_{t-2}}(\mathcal{M}_0) + 2 \frac{1-\gamma}{8\gamma R_{\max}} \epsilon^2$$

after t iterations $\leftarrow$

$$\geq V^{\pi_0}(\mathcal{M}_0) + t \frac{1-\gamma}{8\gamma R_{\max}} \epsilon^2$$

On the other hand, we know that

$$0 \leq V^{\pi}(\mathcal{M}_0) \leq \frac{R_{\max}}{1-\gamma}.$$

$$0 \leq R(s,a) \leq R_{\max}$$

$$\Rightarrow \frac{R_{\max}}{1-\gamma} \geq t \frac{1-\gamma}{8\gamma R_{\max}} \epsilon^2 \Rightarrow t \leq \frac{8\gamma R_{\max}^2}{(1-\gamma)^2 \epsilon^2}$$

$\longrightarrow$ The algorithm runs for at most $\left\lceil \frac{8\gamma R_{\max}^2}{(1-\gamma)^2 \epsilon^2} \right\rceil$.

Now, we characterize the near-optimality of the algorithm.

Performance difference lemma $\nearrow$

$$V_{(\mathcal{M}_0)}^* - V_{(\mathcal{M}_0)}^{\pi_T} = \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi^*}} \left[\mathbb{E}_{a \sim \pi(\cdot|s)} [A^{\pi_T}(s,a)] \right]$$

arg $\leq \max \leftarrow$

$$\leq \frac{1}{1-\gamma} \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi^*}} \left[\max_{a \in \mathcal{A}} A^{\pi_T}(s,a) \right]$$

$$\leq \frac{1}{1-\gamma} \left(\max_{s \in \mathcal{S}} \frac{\bar{\rho}_{\mathcal{M}_0}^{\pi^*}(s)}{\bar{\rho}_{\mathcal{M}_0}^{\pi_T}(s)} \right) \mathbb{E}_{s \sim \bar{\rho}_{\mathcal{M}_0}^{\pi_T}} \left[\max_{a \in \mathcal{A}} A^{\pi_T}(s,a) \right]$$

$q(x) \neq 0 \nearrow$

$$\left[\begin{aligned} \mathbb{E}_{x \sim p} [f(x)] &= \sum_{x \in \mathcal{X}} p(x) f(x) = \sum_{x \in \mathcal{X}} \frac{p(x)}{q(x)} q(x) f(x) \\ &\leq \left(\max_{x \in \mathcal{X}} \frac{p(x)}{q(x)} \right) \sum_{x \in \mathcal{X}} q(x) f(x) = \left(\max_{x \in \mathcal{X}} \frac{p(x)}{q(x)} \right) \mathbb{E}_{x \sim q} [f(x)] \end{aligned} \right]$$

$$\leq \frac{1}{(1-\gamma)^2} \left(\max_{s \in S} \frac{\bar{\rho}_{\mu_0}^{\pi^*}(s)}{\mu_0(s)} \right) \mathbb{E}_{s \sim \bar{\rho}_{\mu_0}^{\pi_T}} \left[\max_{a \in A} A^{\pi_T}(s, a) \right]$$

$$\left[\rho = \mu_0 + \gamma (\rho^\pi)^\top \rho \implies \rho(s) \geq \mu_0(s) \implies \bar{\rho}(s) = (1-\gamma) \rho(s) \geq (1-\gamma) \mu_0(s) \right]$$

termination criterion $\leftarrow \leq \frac{\varepsilon}{(1-\gamma)^2} \left(\max_{s \in S} \frac{\bar{\rho}_{\mu_0}^{\pi^*}(s)}{\mu_0(s)} \right) \cdot$



Note: The guaranteed monotonic improvement and the performance guarantees of the CPI algorithm can be extended to the setting with imperfect, but effective supervised learning.